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A framework for PV yield-based solar energy mapping in mid-low latitude regions with complex topography

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ABSTRACT

Accurate solar energy potential mapping is crucial for effective energy planning, particularly in regions with complex terrain and diverse microclimates. This study presents a novel framework that integrates lapse-rate-adjusted ERA5-Land reanalysis data with satellite-derived solar irradiance to construct a regionally calibrated Gridded Typical Meteorological Year (GTMY) dataset at 2 km resolution. Unlike methods using only ground data or coarse reanalysis, this approach applies topographic lapse rate corrections to improve temperature and dew point accuracy by 25.5% and 16.7%, while also assessing wind speed bias. The GTMY is validated across 18 ground stations in a geographically compact yet topographically diverse subtropical island region with coastal plains, and mountainous terrain, providing rigorous testbed representative of many mid- and low-latitude regions worldwide. Results show high spatial and temporal agreement with long-term observations, with an annual average GHI difference of +14 kWh/m^2. PV performance modeling based on GTMY inputs yields spatially realistic solar energy yield maps. The demonstrated robustness of this framework under heterogeneous conditions highlights its broad applicability to other regions with complex geophysical characteristics. This research provides a scalable and transferable solution for accurate solar resource assessment and supports adaptive infrastructure planning under both current and future climate conditions.

1. Introduction

The global shift toward renewable energy has heightened the demand for accurate solar resource assessment tools, with solar energy emerging as a pivotal element in sustainable power systems. Advances in photovoltaic (PV) technology and declining installation costs have accelerated solar deployment across diverse geographic settings. However, accurately mapping solar energy potential remains challenging—particularly in mid-low latitude regions with complex terrain and microclimates [1]. Conventional approaches, often based on ground-based Typical Meteorological Year (TMY) datasets, are limited by sparse station coverage, calibration issues, and an inability to capture mesoscale variability. Ground-based observations often lack sufficient spatial representativeness, with these limitations manifesting particularly in the prohibitively high operational costs of maintaining solar radiation monitoring networks, resulting in many regions having only sparse coverage of Automatic Solar Radiation Stations across vast territorial expanses [2]. Especially, in topographically complex regions, spatially sparse stations cannot capture the variability of solar radiation across surrounding terrain [3]. Moreover, temporal gaps and irregularities in ground data can smooth important short-term variations [4]. This reduces the reliability of monthly averaged solar values, which are critical for energy planning. In mountainous and coastal areas, solar irradiance is affected by terrain-induced shading, atmospheric dynamics, and localized climate patterns [5]. These factors introduce significant spatial and temporal heterogeneity, rendering generalized assessments insufficient for local-scale energy planning. Accurately characterizing these variations requires refined modeling frameworks capable of integrating high-resolution topographical and meteorological data [6].

TMY datasets, while useful for representing average climatic conditions, often fail to capture inter-annual variability and extreme events—factors critical for evaluating long-term solar energy performance and financial risk. Their spatial resolution is inherently limited by the distribution of weather stations, particularly in remote or mountainous areas, where sharp gradients in solar radiation are common. The point-to-grid representativeness mismatch creates substantial verification discrepancies, especially in topographically complex regions where wind patterns and radiation fields exhibit spatial gradients within distances shorter than typical station spacing [7]. Interpolation of point-based measurements in such domains introduces substantial uncertainty [8].

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Additionally, data quality concerns — ranging from inconsistent calibration to operational gaps — further diminish their reliability for high-resolution solar mapping [9]. As alternatives, global atmospheric reanalysis products provide globally consistent, high-temporal-resolution meteorological data by assimilating satellite and ground-based observations. Their broad spatial coverage and accessibility make them attractive for solar resource assessments in data-scarce regions.

However, their spatial resolution remains coarse which makes them directly inadequate for local-scale solar assessments, especially in topographically complex areas [10]. This coarse resolution means that fine-scale terrain variability cannot be captured, as diverse surface and atmospheric features are averaged within each grid cell [11]. This averaging process introduces significant biases in solar radiation estimates. For example, ERA5 exhibits systematic biases in solar irradiance estimates, with both over and underestimations reported depending on cloud dynamics, aerosol representation, and regional climatic conditions [7]. Cloud representation is a particularly critical weakness in reanalysis models. Under cloudy conditions, these models tend to overestimate solar radiation. In contrast, they often underestimate radiation during clear-sky periods. This is largely due to inadequate representation of cloud fraction and optical properties. In mountainous regions, accurately capturing terrain-driven variables such as temperature [12] and wind speed [13], both of which influence PV output, is challenging due to elevation mismatches and coarse horizontal resolution. As solar energy planning extends to diverse and topographically complex environments under climate change pressures, the need for dynamically calibrated, high-resolution datasets has become increasingly critical.

To overcome these limitations, this study proposes a novel framework that integrates topographically corrected ERA5-Land reanalysis with satellite-derived irradiance estimates to construct a regionally calibrated Gridded Typical Meteorological Year (GTMY) dataset. Lapse-rate adjustments are applied to temperature and dew point temperature using a high-resolution digital elevation model (DEM), mitigating elevation-induced biases and improving the representation of near-surface atmospheric conditions. A regionally adapted Heliosat −2 model is then used to incorporate satellite-derived irradiance, resulting in a 2 km resolution GTMY tailored to mid-low latitude regions with complex terrain. Unlike conventional studies that rely solely on Global Horizontal Irradiance (GHI) [1], our framework emphasizes PV power yield as the primary metric for solar potential mapping. PV yield, expressed in kWh, accounts also for dynamic performance factors such as temperature, wind effects, and system losses, thereby providing a more realistic estimate of electricity generation potential under real-world operating conditions. This study aims to develop and validate a high-resolution framework for solar energy potential mapping in mid-low latitude regions with complex terrain, using a regionally calibrated GTMY dataset. The specific objectives are to:

1. Construct a 2 km GTMY dataset by integrating lapse-rate corrected ERA5-Land reanalysis and satellite-derived irradiance;
2. Apply topographically informed lapse rate corrections to enhance temperature and humidity accuracy;
3. Assess and correct biases in ERA5-Land variables through comparison with ground station observations;
4. Map solar potential using both Global Horizontal Irradiance (GHI) and PV power yield metrics.

By addressing these goals, this research advances solar resource assessment methodologies for geographically complex regions and provides a scalable, transferable tool for energy system planning. The use of PV power yield offers a more accurate representation of actual energy generation, supporting site selection, investment strategies, and long-term climate adaptation planning.

2. Methodology

The methodology adopted in this study integrates satellite-derived irradiance, reanalysis-based meteorological data, and photovoltaic (PV) performance modeling to construct a regionally calibrated Gridded Typical Meteorological Year (GTMY) dataset and assess solar energy potential. Fig. 1 illustrates the overall workflow. The methodology is organized into three components: (1) satellite-based irradiance estimation and separation, (2) GTMY dataset construction through reanalysis correction and typical month selection, and (3) PV yield modeling and potential classification.

2.1. Satellite-derived solar irradiance and separation modeling

GHI was estimated using the Heliosat −2 model, a statistical method that derives GHI from satellite imagery on a pixel-by-pixel basis from a visible wavelength spectrum. The method originates from the work of Cano et al. [14], who introduced the concept of estimating surface irradiance by comparing satellite-observed reflectance to what would be expected under clear-sky conditions. This approach was later refined by Rigollier et al. [15], who developed Heliosat −2 by replacing raw satellite digital counts with calibrated radiance values. This change not only improved accuracy but also enabled to use of data from various satellites.

In this study, a site-adapted version, optimized for mid-low latitude regions [16], was applied. This version has been previously validated for the study area. However, due to a lack of reliable station data, additional evaluation of Heliosat −2 model in this research was not possible. There were noticeable inconsistencies in GHI measurements before and after the calibration of pyranometer sensors in the station dataset described later. Errors were particularly evident before 2003, as noted by Lin and Liu [17], with significant changes later also observed after each calibration period.

The Heliosat −2 model begins by normalizing reflectivity by calculating apparent albedo ρ_t at a given time as follows Eq. (1):

$$\rho_t = \frac{\text{Reflectivity}}{\epsilon \cdot \cos(\theta_s)} \tag{1}$$

where reflectivity is the value of MTSAT-2 visible band or Himawari Band-3 derived from radiance, ϵ is Sun–Earth distance and θ_s is the solar zenith angle.

The next step is to calculate cloud index n Eq. (2):

$$n = \frac{\rho_t - \rho_g}{\rho_c - \rho_g} \tag{2}$$

where ρ_g and ρ_c represent ground albedo and cloud albedo. Both of these values are taken on a monthly basis. Ground albedo is taken as the second minimum and cloud albedo is taken as 95th percentile value in a month time series.

Later, the clear-sky index K_c is calculated after the cloud index is defined per each time step. This study used the adjusted formula as presented in Han and Vohnicky [16]. This mutual relationship is defined in four partitions as follows Eq. (3):

$$K_c = \begin{cases} 1.2 & \text{if } n \le -0.2 \\ 0.9673 - 1.0242n & \text{if } -0.2 < n < 0.8 \\ 0.9348 - 1.2528n + 0.4355n^2 & \text{if } 0.8 < n \le 1.2 \\ 0.05 & \text{if } 1.2 < n \end{cases} \tag{3}$$

Next, the GHI is obtained by multiplying the clear-sky index with the value of clear-sky irradiance G_{ch} delivered by the REST2 clear-sky model [18], as below Eq. (4):

$$GHI = K_c \cdot G_{ch} \tag{4}$$

Lastly, the GHI values were corrected for systematic error by using the linear subtraction of fitted lines in the scatter plot, as described in Han and Vohnicky [16].

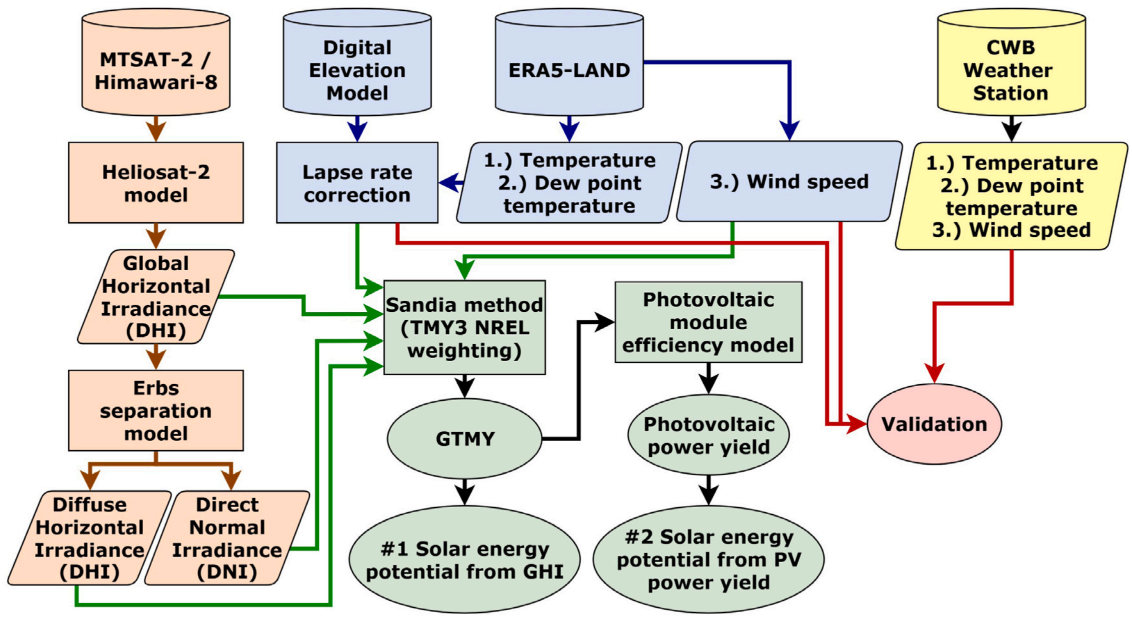

Fig. 1. Flowchart of the implemented methodology.

Note that, the Heliosat −2 model in this study was supplied by data of geostationary weather satellites from MTSAT-2 for years 2010 to 2015 and Himawari-8 for years 2016 to 2021. Post-processed data of MTSAT-2 visible band were downloaded from MCFETCH, web-based API of the University of Wisconsin-Madison Space Science and Engineering Center (http://www.ssec.wisc.edu/). Data of the Himawari-8 band 3 were downloaded from the P-Tree System through the FTP site provided by JAXA (ftp.ptree.java.jp). The clear-sky model REST2 used data delivered from Modern-Era Retrospective Analysis for Research and Applications MERRA-2 [19].

The DNI and DHI are separated from the GHI based on various parameters and conditions. There are two different separation model groups, where one is used to estimate DNI and second is used to estimate DHI. The second of these variables is then calculated based on general formula of GHI, as shown in Eq. (5).

$$GHI = DNI \cdot \cos(\theta_s) + DHI \tag{5}$$

In this study, the Erbs separation model was used to obtain DHI from the satellite-derived GHI estimation [20]. The model established a relationship between the hourly diffuse fraction (df) and the hourly clearness index (k) based on the measurements from the four locations in the United States. This model was selected due its simplicity as it relies only on a single parameters and good results in previously conducted research in mid-low latitude region [21]. The parameter k_t is an indication of the relative clearness of the atmosphere. In general, when the atmosphere is clearer, a smaller fraction of the irradiance is scattered. The parameter k_t therefore, can be calculated based on the following Eq. (6):

$$k_t = \frac{GHI}{I_0 \cdot \cos\theta_s} \tag{6}$$

where I_0 is the extraterrestrial irradiance and θ_s is the solar zenith angle.

The diffuse fraction ratio of hourly diffuse radiation can be expressed as a relationship of DHI/GHI. Thus, in order to obtain the DHI, we firstly must calculate the term df as proposed by Erbs, here in Eq. (7).

$$df = \begin{cases} 1.0 - 0.09k_t & \text{for } k_t \leq 0.22 \\ 0.9511 - 0.1604k_t + 4.388k_t^2 - 16.638k_t^3 + 12.336k_t^4 & \text{for } 0.22 < k_t \leq 0.8 \\ 0.165 & \text{for } k_t > 0.8 \end{cases} \tag{7}$$

DHI was later calculated by Eq. (8):

$$DHI = GHI \cdot df \tag{8}$$

Eventually, the second part of the GHI, in this case it was the DNI, is calculated based on the expression of Eq. (9) for DNI, as shown in Eq. (9).

$$DNI = (GHI + DHI)/cos(\theta_s) \tag{9}$$

2.2. Generation of GTMY using reanalysis data and TMY selection

To construct a representative climatic dataset at high spatial resolution, a GTMY was generated using ERA5-Land reanalysis data along with satellite derived GHI, DNI, and DHI. A topographically informed lapse-rate correction was applied to temperature, dew point, and wind speed fields based on digital elevation differences. The TMY was then synthesized using the Sandia method.

The Sandia method is an empirical approach that selects individual months from different years of the period of record to construct a TMY [22]. Additionally, note that there must be at least 10 years of data in order to establish TMY that will represent the weather condition in the region. The Sandia method selects a typical month based on daily indices consisting of the maximum, minimum, and mean dry bulb and dew point temperatures; the maximum and mean wind velocity; and the total GHI. Final selection of a month includes consideration of the monthly mean and median and the persistence of weather patterns. The process may be considered a series of 4 steps. As this method is well established, we include detailed description of steps in Appendix A.

Lapse-rate adjustments were applied using the standard environmental lapse rate for temperature and derived rates for humidity. This correction improves the vertical accuracy of ERA5-Land fields in complex terrain.

2.3. PV yield simulation and solar energy potential mapping

Based on the GTMY inputs, PV system performance was simulated across the study area to estimate actual energy generation potential. The specific PV yield was computed for each location using standardized configuration parameters.

To better understand the influence of temperature and wind speed on the potential annual yield of solar PV power, an artificial PV panel was created for each TMY grid point. A single Panasonic HIT panel with a maximum power of $P_{\max} = 325$ W (measurement performed by Sandia National Laboratories [23]) was selected. The tilt and orientation were chosen based on regional findings of Han et al. [24], who determined that the optimal tilt angle θ_T is 13° and the optimal orientation $\theta_{surface}$ is 177°.

Using these values, the total in-plane irradiance I_{total} was calculated using the simple isotropic model Eq. (10):

$$I_{total} = DNI \cdot \cos(\theta_{incidence}) + DHI \cdot \left(\frac{1+\cos(\theta_T)}{2}\right) + \alpha \cdot GHI \cdot \left(\frac{1-\cos(\theta_T)}{2}\right) \quad (10)$$

The angle of incidence $\theta_{incidence}$ was calculated as follows Eq. (11):

$$\theta_{incidence} = \arccos\left(\sin(\theta_s)\cdot\sin(\theta_T) + \cos(\theta_s)\cdot\cos(\theta_T)\cdot\cos(\theta_a - \theta_{surface})\right) \quad (11)$$

where θ_s is the solar zenith angle, θ_T is the tilt angle of the solar panel, θ_a is the solar azimuth angle, and $\theta_{surface}$ is the orientation of the solar panel in the horizontal plane.

Subsequently, the PV panel efficiency was calculated using the ADR model. A detailed description and evaluation of this model for module efficiency is provided in Driesse et al. [25]. The model estimates module efficiency from a function of normalized irradiance (denoted as S) and cell temperature (denoted as T), expressed as $\eta(S,T)$ in Eq. (12):

$$\eta(S,T) = k_a\left[\left(1 + k_{rs} + k_{rsh}\right) v(S,T) - k_{rs}S - k_{rsh}v(S,T)^2\right] \quad (12)$$

where the parameters k_a, k_{rs}, and k_{rsh} are fitting constants that describe the initial conditions, series resistance, and shunt resistance, respectively. It is noteworthy that the ADR model has been demonstrated to have a lower error rate compared to other models used to determine the efficiency of PV modules. For a comprehensive discussion of this model, readers are referred to the relevant literature.

The cell temperature T is calculated based on the method outlined by Faiman [26] as:

$$T = T_a + \frac{I_{total}}{U_0 + U_1 \cdot WS} \quad (13)$$

where the parameters T and T_a are the module and ambient temperatures (°C), U_0 and U_1 are empirical heat transfer coefficients (W/m^{2}K and W/m^{3}sK, respectively), and WS is the wind speed (m/s).

PV power yield is defined as follows:

$$P_{mp} = P_{STC} \cdot \eta(S,T) \cdot \frac{I_{total}}{G_{STC}} \quad (14)$$

where the parameters are defined as follows: $P_{STC} = 325$ W is the rated module power under standard test conditions, $G_{STC} = 1000$ W/m^2 is the irradiance at STC. Finally, the total PV power yield is determined by summing the values from the TMY dataset.

The solar energy potential, based on the TMY dataset, was categorized into three distinct categories after statistical normalization of two complementary metrics. Specifically, solar energy maps were created using two metrics: (i) the annual sum of GHI, and (ii) the total specific PV power yield. Both metrics were normalized using z-score standardization: (value - mean)/standard deviation. This statistical approach provides a more objective and reproducible classification, especially for regional studies, compared to subjective methods such as natural breaks (Jenks) classification used in previous solar potential studies (Prăvălie et al. [27]). The ±1σ boundaries represent statistically significant deviations from the regional mean, where σ represents one standard deviation. It should be noted that the ±1σ thresholds do not have inherent physical meaning but rather represent a statistically-based approach to identify areas that deviate significantly from the regional average solar resource distribution. This relative classification is particularly appropriate for regional solar energy planning, as it identifies locations with comparatively better or worse solar potential within the study area. After normalizing these values, three categories were established: the "poor" category for values below -1σ, the "fair" category for values between -1σ and $+1\sigma$, and the "good" category for values above $+1\sigma$. These categories provide a clear classification of solar energy potential across the region, reflecting varying levels of suitability for solar energy generation.

2.4. Matrices for reanalysis evaluation

The accuracy analysis of ERA5-Land reanalysis is based on two common metrics: the mean bias deviation and root mean square deviation. The equations of these metrics are as follows Eqs. (15) and (16):

$$MBD = \frac{\sum_{i=1}^{N}(x_i - y_i)}{N} \quad (15)$$

$$RMSD = \sqrt{\frac{\sum_{i=1}^{N}(x_i - y_i)^2}{N}} \quad (16)$$

where x stands for station observation data, y stands for ERA5-Land corresponding pixel at the same place as station and i is hourly based index.

3. Data sources

Taiwan was selected as the validation area due to its diverse geographical and climatic conditions, spanning from tropical (south) to temperate (north) climates while being bisected by the Tropic of Cancer. Its varied terrain — western plains and eastern forest-covered mountain ranges with over 200 peaks exceeding 3000 m (maximum Yushan, 3952 m) — within a 36,197 km^2 area makes it ideal for testing GTMY dataset performance across different environments. A 20 m resolution digital elevation model from Taiwan's open data portal (https://data.gov.tw) was downscaled to 2 km to match solar irradiance data resolution (Figure B.1). All data described below covers the 12-year period from 2010 to 2021.

3.1. Reanalysis data

ERA5-Land provides reanalysis data at 9 km grid resolution with hourly temporal resolution [28], covering only land areas defined by a land–sea mask [29]. In the presented study, the ERA5-Land 2 m air temperature, 2 m dew point temperature, and 10 m u and v components of wind were collected from ECMWF from the Climate Data Store application (https://cds.climate.copernicus.eu/).

The wind direction increase clockwise from North, therefore, based on basic trigonometry, the wind velocity V was calculated as Eq. (17):

$$|V| = \sqrt{u^2 + v^2} \quad (17)$$

The air and dew temperature obtained from ERA5-Land were upscale based on Taiwan DEM to 2 km spatial resolution based on lapse rate correction. To do so, first the geometric height of ERA5-Land was approximated based on geopotential height stored as geopotential in $\frac{m^2}{s^2}$ in original reanalysis dataset by following Eq. (18):

$$alt = \frac{Re^2}{Re - h} \quad (18)$$

where Re is the radius of the Earth (6,371,008.8 m) and h is the geopotential height calculated from geopotential by dividing it by fixed Earth's gravitational acceleration of 9.80665 $\frac{m^2}{s^2}$. The final digital elevation model of ERA-5 Land used in this study is shown at Figure B.2.

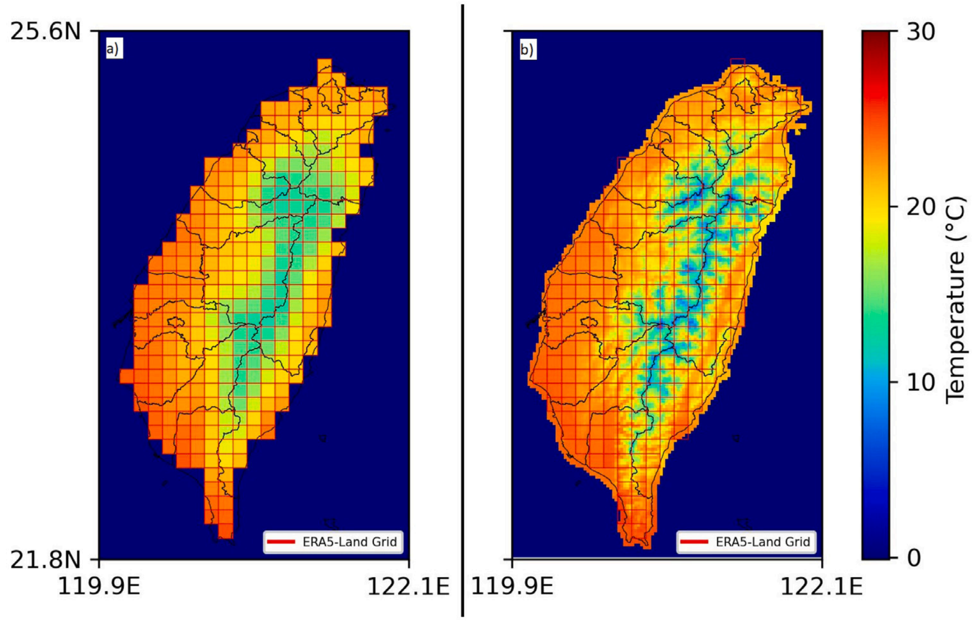

Fig. 2. (a) Annual mean temperature of ERA5-Land, (b) Lapse rate corrected annual mean temperature.

Second, ERA5-Land pixels near the coastlines were extrapolated by taking the average value of the nearest neighbor inside of the 3 × 3 matrix, details in Figure B.3.

Finally, the temperature and dew-point temperature were upscaled by lapse rate L with assumption of temperature decrease of 0.65 °C and dew-point temperature decrease of 0.2 °C per 100 m difference [30]. This corresponds to the following Eq. (19):

$$T = \frac{(T_{\mathrm{ERA5}} - L \times (\mathrm{alt} - \mathrm{alt}_{\mathrm{ERA5}}))}{100} \tag{19}$$

where T is the lapse corrected temperature, T_{ERA5} is the temperature from ERA5-Land, alt is the grid elevation in Taiwanese DEM and alt_{ERA5} is the grid elevation of previously calculated DEM of ERA5-Land. Result of upscaled temperature represented as the annual mean temperature is shown in Fig. 2.

3.2. Ground station measurement data

The Central Weather Administration (CWA) is the government meteorological research and forecasting institution in Taiwan, and it is also in charge of collecting data from weather stations. This study used data from 18 weather stations across the island, shown in Fig. 3. The temperature and dew-points were collected in the hourly average value and wind speed is the average of 10 min before the hour. The weather stations were split into 3 different groups due to their location regarding the ERA-5 Land grid and their altitude. The stations in the Group A are located inside of the ERA-5 Land, the stations in the Group B are inside of the extrapolated ERA-5 Land grid and Group C stations have elevation more than 1000 m above the sea level. The detailed description of stations location and altitude is written in Table B.1.

4. Results

This section presents the GTMY framework evaluation results in three parts: (1) reanalysis data validation; (2) satellite irradiance and GTMY representativeness; and (3) solar energy potential assessment.

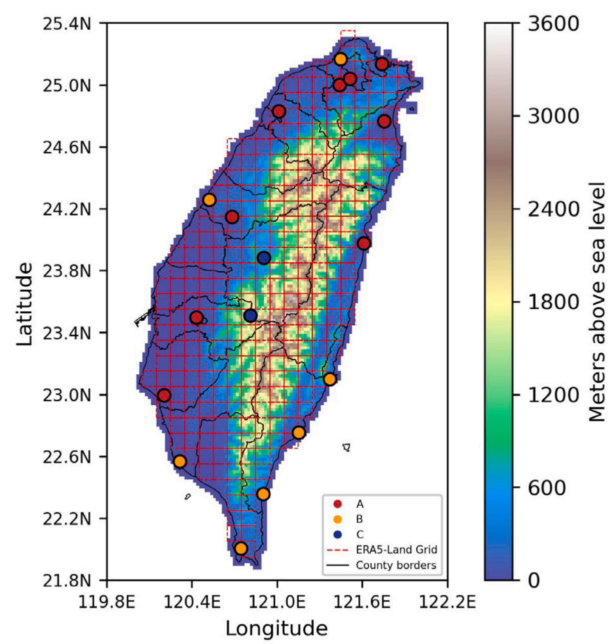

Fig. 3. The weather stations locations regarding their elevation and ERA-5 Land grid.

4.1. Validation and spatiotemporal patterns of satellite-derived solar irradiance

Fig. 4 shows satellite-derived GHI, DNI, and DHI data for Taipei (2010–2021). The data exhibits significant daily and seasonal variability with summer peaks reaching approximately 1000 W/m^2 hourly peaks. Monthly GHI means remain relatively consistent throughout the observed period. DNI shows less clear seasonality with higher peaks than GHI and a noticeable shift after 2015 when the new Himawari-8 satellite became operational. DHI shows lower values as it captures only scattered radiation.

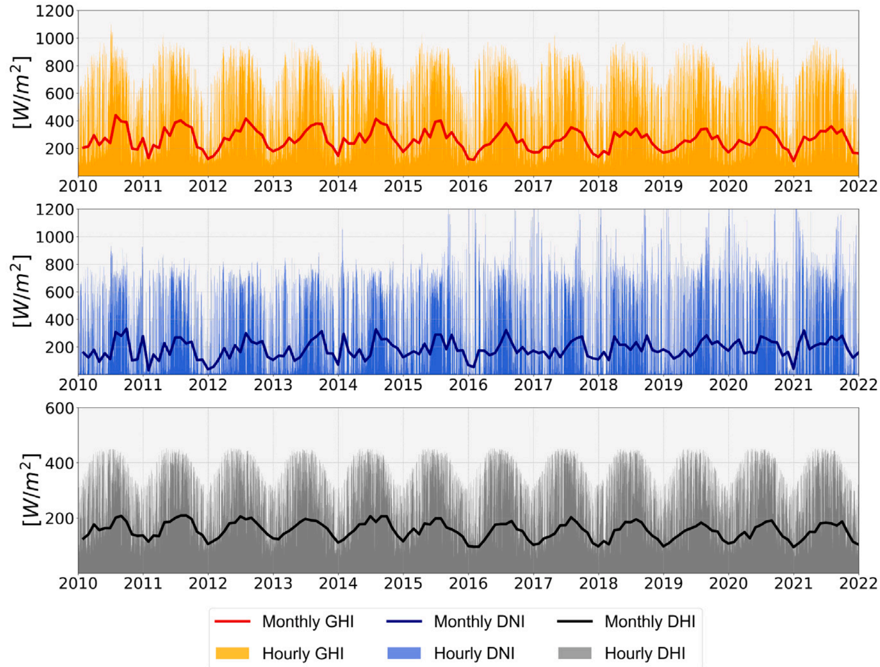

Fig. 4. Satellite derived GHI, DNI and DHI data at Taipei station.

Details of the long-term annual sum of GHI for Taiwan are shown in Figure B.4 in the appendix, measured in kilowatt-hours per square meter (kWh/m^2). Southern regions display optimal conditions for solar energy production with values exceeding 1700 kW per year, while northern and some central regions show lower potential. Orographic effects are evident with the east coast receiving less GHI due to cloudier conditions.

4.2. Reanalysis correction and GTMY evaluation

The MBD and RMSD box plots displayed in Figs. 5 and 6 provide an evaluation of the ERA5-Land reanalysis temperature data accuracy across different groups of weather stations. The MBD plot reveals that biases vary throughout the year for all groups, with noticeable differences between the original data (A, B, C) and the corrected data (A_cor, B_cor, C_cor). Group A (light blue) exhibits relatively consistent MBD values with slight positive biases, while the corrected data (A_cor, dark blue) shows reduced biases, indicating an improvement in temperature estimates by simple lapse rate correction. Group B (light green) displays larger positive biases with larger interquartile range during autumn and winter months. Though, after correction (B_cor, dark green), these biases are significantly reduced, especially during the summer months. Group C (pink), representing high-altitude stations, shows the highest negative MBD, which is notably reduced after correction (C_cor, red), though some bias remains.

The RMSD box plot indicates the overall deviation magnitude, with Group A showing moderate RMSD values and improved accuracy after lapse rate correction. Group B, with higher RMSD values, sees notable reductions. Group C has the highest RMSD values, reflecting substantial deviations in high-altitude regions, but significant improvements are seen in the corrected data, especially from spread perspective. Seasonal variation analysis shows that winter months exhibit higher MBD and RMSD values, especially for Group C, indicating greater variability and errors in colder months. Overall, lapse rate correction improves accuracy across all station groups:

- Group A: MBD improved from 1.66 to 1.27; RMSD from 2.22 to 1.96
- Group B: MBD improved from 2.13 to 1.61; RMSD from 2.66 to 2.24
- Group C: MBD improved from −2.8 to 1.2; RMSD from 3.02 to 2.25

Seasonal variation analysis reveals higher errors in winter months, particularly for Group C (high-altitude stations).

ERA5-Land dew temperature evaluation's box plots on Figs. 7 and 8 present a comparison of hourly dew point temperature data between ERA5-Land and station data, categorized into three groups based on location and altitude. Group A, which includes stations located inside the ERA5-Land grid, shows MBD values that fluctuate around zero throughout the year. This indicates that ERA5-Land data accurately represents temperatures at these locations. RMSD values for Group A remain low, around 2 °C, demonstrating good agreement with minor errors. Small improvement can be seen in the lapse rate corrected A_cor; thought, this improvement is much smaller than at temperature dataset. Group B with stations inside the extrapolated ERA5-Land grid, also has MBD values closely around zero. There is slightly higher variability compare to Group A. The RMSD values for Group B are slightly higher than at Group A, indicating reasonable accuracy with marginally larger errors. The lapse rate corrected version B_cor improved results but once again, this improvement was much smaller than in temperature. Finally, Group C shows consistently negative MBD values, suggesting that ERA5-Land tends to overestimate dew temperatures for higher altitude stations. Unfortunately, not much improvement can be seen by lapse rate correction in dataset of group C_cor. The RMSD values for Group C are significantly higher, often reaching up to 6-8 °C in certain winter months, reflecting a larger discrepancy between ERA5-Land data and station data at high altitudes. It can be observed that lapse rate corrected version C_cor was able decrease these large errors during whole year, pointing out that lapse rate correction can still be effective. Overall, Group A demonstrates the best accuracy and consistency, with the average annual MBD (based on monthly medians) improving and RMSD decreasing slightly.

- Group A: MBD improved from 0.1 to −0.02; RMSD from 1.44 to 1.42

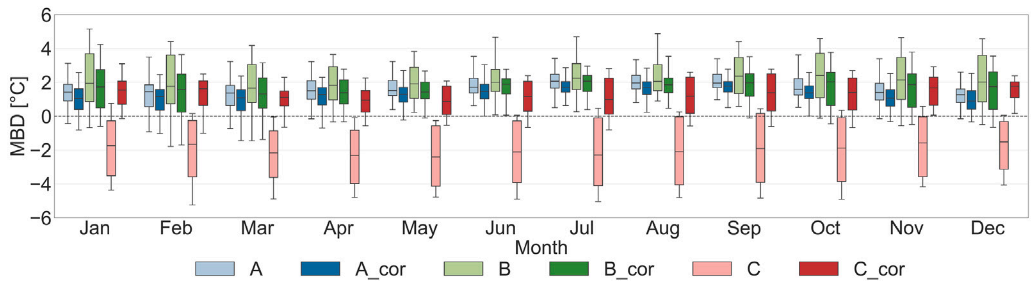

Fig. 5. Hourly MBD of temperature per each month.

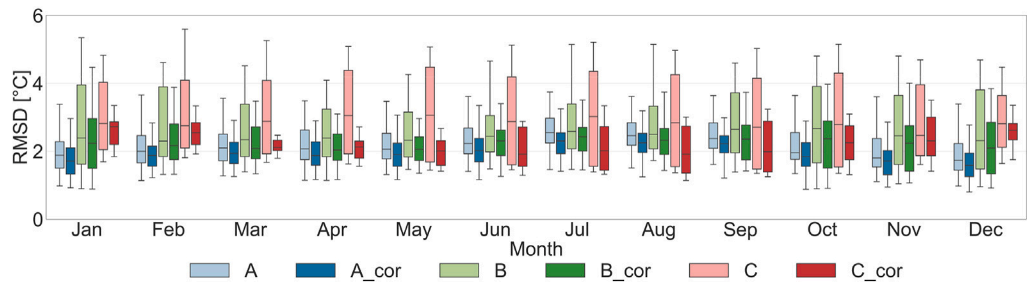

Fig. 6. Hourly RMSD of temperature data per each month.

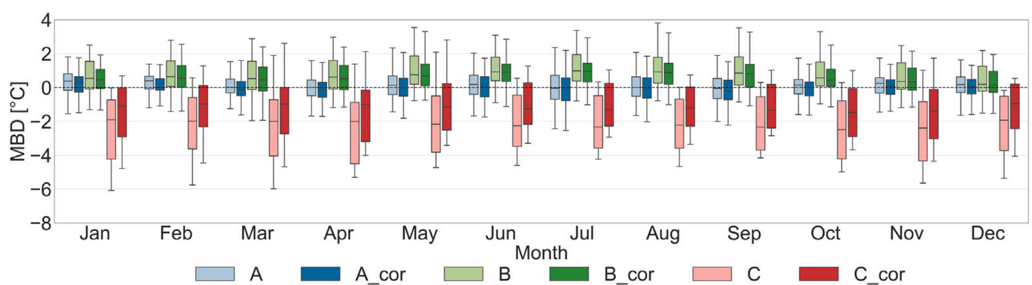

Fig. 7. Hourly MBD of dew point temperature per each month.

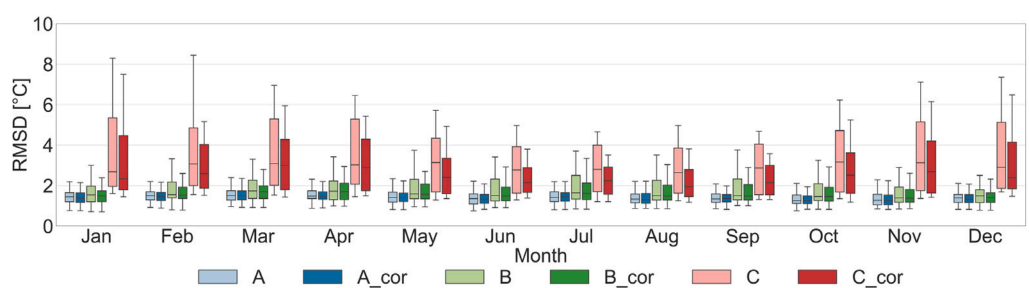

Fig. 8. Hourly RMSD of dew point temperature per each month.

- Group B: MBD improved from 0.8 to 0.64; RMSD from 1.74 to 1.64
- Group C: MBD improved from −2.25 to −1.24; RMSD from 3.3 to 2.75

The MBD and RMSD box plots in Fig. 9 provide an evaluation of ERA5-Land reanalysis wind speed data accuracy across different weather station groups. Group A generally shows negative MBD, indicating ERA5-Land overestimates wind speed, with RMSD values around 1 to 2 m/s. Group B exhibits larger negative MBD, especially in spring and autumn, and higher RMSD values often exceeding 2 m/s. Furthermore, the large interquartile variability can be seen at group B, especially during autumn and winter seasons. Group C, representing high-altitude stations, shows smaller MBD close to zero and the lowest RMSD values, typically below 1 m/s, indicating better agreement with ERA5-Land data. Overall, ERA5-Land data reasonably represents wind speeds at high altitudes (Group C) but tends to overestimate them at lower altitudes (Groups A and B). Details on MBD and RMDS of each all stations are in Table B.2. Group A exhibiting mixed biases, Group B showing higher variability due to coastal locations, and Group C (high-altitude stations) having smaller RMSD values indicating better agreement with ERA5-Land data. This variability across stations underscores the influence of local climate, orographic, and coastal parameters on wind speed measurements, as ERA5-Land's large grid size struggles to capture coastal influences, leading to inconsistent results. Although

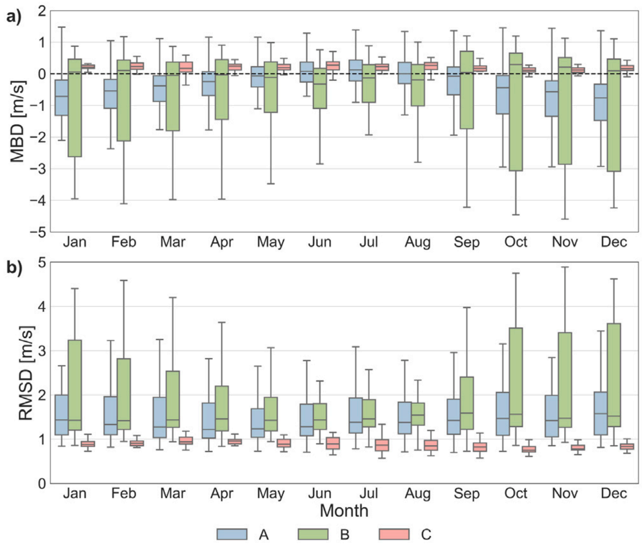

Fig. 9. (a) Hourly MBD and (b) Hourly RMSD of wind speed per each month.

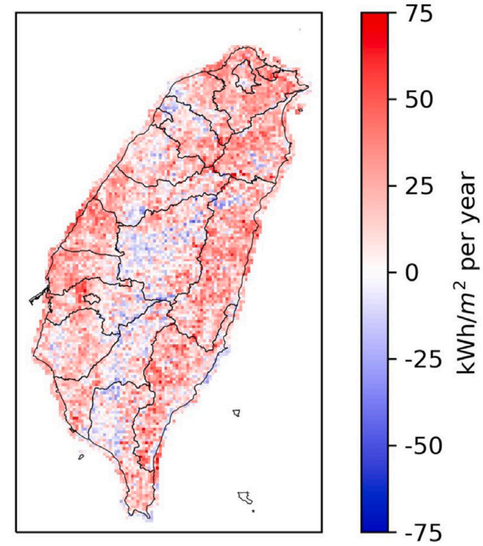

Fig. 10. Difference of annual sum GHI between TMY and long-term average.

the overall bias remains minimal, highlighting ERA5-Land's suitability for this study, the results vary slightly across groups.

- Group A: MBD = −0.46; RMSD = 1.61
- Group B: MBD = −0.68; RMSD = 1.95
- Group C: MBD = 0.19; RMSD = 0.88

These values represent the average annual MBD and RMSD, based on the median values of each month. ERA5-Land generally overestimates wind speed at lower altitudes (Groups A and B) but shows better agreement at high altitudes (Group C). Details for individual stations are in Table B.2.

The performance of reanalysis data after lapse-rate correction provides a reliable foundation for constructing the GTMY dataset. The following results compare the GTMY-derived climate variables with their long-term climatological means to assess regional representativeness.

Fig. 10 shows the differences in GHI between the GTMY and long-term average for Taiwan. The map reveals that most regions in Taiwan exhibit higher GHI values in the GTMY compared to the long-term average, as indicated by the extensive red areas covering the majority of the island. Only a few regions, particularly in the central part and from there going south, show lower GHI values in the GTMY. Results suggest that the GTMY generally predicts more solar energy availability than the long-term average across most of Taiwan but lower in the mountainous area.

Figures B.5, B.6, B.7 show the differences in mean and standard deviation statistics between the GTMY and long-term averages for Taiwan island in temperature, dew point temperature, and wind speed, respectively. The temperature map reveals that GTMY temperatures are generally higher in the northern and eastern regions compared to the long-term average. For dew point temperature, the GTMY dataset shows higher values in most regions, with exceptions in the south and central areas. Wind speed differences indicate that the central eastern region has lower wind speeds in the GTMY, while the southern and western coasts show higher wind speeds compared to the long-term average. There was no significant difference in standard deviations between GTMY and long-term averages.

These results demonstrate that topographically-informed lapse-rate correction significantly improves the reliability of reanalysis-based GTMY datasets, especially in high-altitude and extrapolated regions.

4.3. PV power yield and solar energy potential

The Fig. 11 categorizes solar energy potential into three categories based on GHI values: "poor" (below -σ), "fair" (between -σ and +σ), and "good" (above +σ). The GHI values at -σ and +σ were 1238 kWh/m^2 and 1504 kWh/m^2, respectively, with regional mean of 1371 kWh/m^2. The counts of grids in each category based on GHI were as follows: 1608 in the poor category, 5881 in the fair category, and 1429 in the good category. For cumulative PV power output, the corresponding thresholds were 373 kW and 461 kW, with a regional mean of 417 kW. Similarly, the PV power yield count in these categories were 1609, 5840, and 1469, respectively. Thus, the differences in counts between the two different approaches were very small. The solar potential map based on PV power yield (Fig. 11, panel b) shows a higher number of grids in the good category, as it takes temperature and wind speed into account as factors influencing the solar energy potential. This change is particularly evident in the central-west region, where grids from the fair category move into the good category. On the other hand, in the usually warmer south-west region, higher temperatures lead to a decrease in solar potential, causing grids to shift from the good category to the fair category. The rest of the island shows almost identical results across both categories.

The PV power yield mapping reflects the real-world effects of ambient conditions on solar generation potential and provides a more realistic foundation for infrastructure planning compared to traditional GHI-based estimates. To further examine the sensitivity of PV power yield to atmospheric lapse rate assumptions, we conducted a comparative analysis using TMY datasets with and without lapse rate correction. The results, presented in Figure B.8, show that elevation positively influences PV output under the lapse rate-adjusted scenario, as evidenced by a negative regression slope which indicates that incorporating lapse rate correction tends to increase PV yield estimates at higher elevations.

5. Discussion

This study demonstrates the effectiveness of our integrated approach in improving solar resource assessment for regions with complex topography. Our evaluation of ERA5-Land reveals systematic temperature biases consistent with previous research [31,32], where temperatures are underestimated at lower altitudes and overestimated

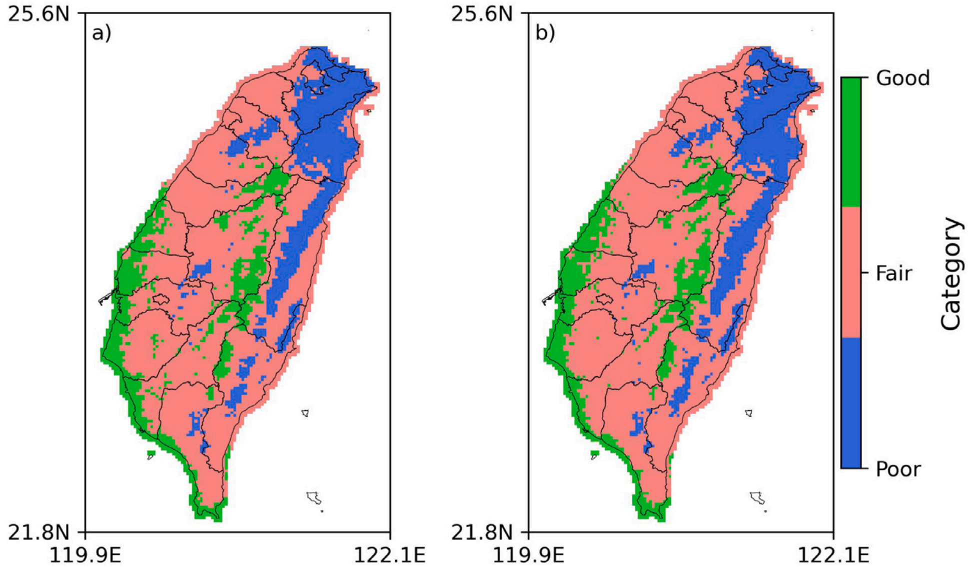

Fig. 11. Solar energy potential based on the TMY dataset as (a) the annual sum of GHI, and (b) the total PV power yield.

at higher elevations. Unlike traditional methods, our implementation of topographically-informed lapse rate correction significantly reduces this bias across elevation ranges, extending the findings of Negm et al. [33] to more diverse landscapes. The correction's effectiveness diminishes in extrapolated coastal grids, supporting Pelosi et al. [34], which highlights an area for methodological advancement particularly relevant for island nations.

The dew point temperature analysis further validates our approach, showing reliable performance in lowlands with modest improvements through lapse rate correction. However, the persistent larger biases at high elevations indicate a fundamental limitation of ERA5-Land that our methodology partially addresses. This gradient of reliability with elevation represents an important consideration for solar resource mapping in mountainous regions.

Our wind speed evaluation diverges from Vanella et al. [32], revealing overestimation in group A stations rather than underestimation. This finding aligns with Arregocés et al. [35] and Khadka et al. [36], suggesting ERA5-Land's tendency to overestimate wind speed near coastlines and at higher elevations. This insight informed our integrated approach, where we incorporated these biases into our solar potential calculations, particularly for PV power yield estimates where wind speed affects panel temperature and efficiency.

The Heliosat −2 method implementation represents another advancement of our framework, successfully capturing intraday GHI variability [16,37]. By integrating satellite data with corrected reanalysis, our approach effectively models regional variations in solar potential, including the pronounced north–south gradient visible in Figure B.4. This pattern correlates with precipitation distribution [38,39], demonstrating how our methodology captures complex climate–topography interactions that affect solar resource availability [40].

The GTMY generation represents a significant methodological contribution, producing temperature values that closely match long-term means while preserving realistic variability in dew point and wind speed, consistent with Pusat et al. [41] and Putra et al. [42]. The discrepancy between GTMY and long-term averages stems from inherent limitations in the statistical selection method, which prioritizes certain meteorological parameters over others during the "most representative" month selection process. This weighting approach, while reflecting specific application needs, may result in deviations from long-term means for non-prioritized variables [43], particularly in regions with high interannual variability or complex terrain. In such areas, extreme years can significantly influence the typical year construction [44].

The observed $-\sigma$ and $+\sigma$ GHI values aligned with Prăvălie et al. [27], validating our categorization approach. More importantly, our novel comparison between GHI-based and PV power yield-based solar potential mapping demonstrates that while spatial patterns are similar, incorporating module temperature effects provides crucial additional insights for deployment planning. The higher PV yield after lapse-rate correction, as presented in Fig. B.8, can be explained by the fact that ERA5-Land tends to report higher temperatures than observed at high-elevation grid points. The resulting lower temperatures help prevent overheating of the PV panels, thereby improving their efficiency and increasing PV yield.

This distinction becomes particularly valuable in the context of climate change scenarios. Under RCP8.5, regions in our study area may experience significant increases in hot days ($\geq 30,^{\circ}$ C) by 2050–2060 [45], and potentially shift from temperate to tropical classification [46]. Our PV power yield approach explicitly accounts for these temperature effects on panel efficiency, offering more realistic projections than conventional GHI-based methods. This enhancement demonstrates how our integrated methodology not only improves current solar resource assessment but also provides a more robust foundation for climate adaptation planning.

The GTMY dataset can be effectively applied to enhance the accuracy of long-term PV generation assessments, particularly in evaluating self-use ratio of PV revenue and feasibility of building attached PV systems across diverse urban contexts [47]. Additionally, enhanced data quality and assessment methodologies reduce investment uncertainties, a critical factor demonstrated to attract private capital and accelerate renewable energy financing in support of Paris Agreement objectives [48]. Nevertheless, while our framework already provides high spatial resolution and improved accuracy for PV assessments, it can further support the development of more precise solar resource mapping in complex terrain. For instance, Zorzetto et al. [49] demonstrate that accounting for sub-grid shading and terrain effects recovers significant spatial variability of solar irradiance in mountainous regions, improving the realism of solar radiation models. Additionally, deep learning methods applied to high-resolution digital elevation models can enhance solar energy potential maps [50], enabling precise rooftop solar panel placement, especially in urban areas. Integrating such approaches with our dataset could further refine solar deployment strategies across diverse landscapes.

6. Conclusion

This study developed and validated a novel framework that integrates lapse-rate-adjusted ERA5-Land reanalysis data with satellite-derived solar irradiance to construct a regionally calibrated Gridded Typical Meteorological Year (GTMY) dataset. Validation across a topographically diverse subtropical island, encompassing coastal plains, basins, and mountainous regions, demonstrated strong agreement with long-term observations. This setting provides a rigorous testbed reflective of broader mid- and low-latitude environments, highlighting the applicability of the framework to other regions with complex geophysical characteristics. Evaluation of ERA5-Land reanalysis data revealed systematic temperature biases related to elevation, reasonable dew point accuracy in lowland areas but notable biases at higher altitudes, and general overestimation of wind speeds, particularly in coastal and elevated zones. Lapse-rate correction significantly reduced these biases, providing a reliable foundation for GTMY generation. Furthermore, solar energy potential mapping based on PV power yield improved spatial realism compared to GHI-based assessments by incorporating temperature and wind speed effects on photovoltaic system performance. The demonstrated robustness of this integrated approach supports more accurate solar resource projections and adaptive energy planning under current and future climate conditions. Future work will focus on developing user-friendly assessment tools and building-specific digital twin models to further enhance solar energy deployment strategies and climate adaptation planning.

CRediT authorship contribution statement

Jen-Yu Han: Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization. **Yi-Hsuan Kan:** Writing – original draft, Software, Methodology, Conceptualization. **Petr Vohnicky:** Writing – original draft, Validation, Methodology, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.renene.2025.124512.

Data availability

The data used in this study are subject to specific access conditions. The CWA data are restricted, and readers interested in accessing this dataset should refer to the CWA website (https://www.cwa.gov.tw/eng/) for further details on permissions and availability. In contrast, the GTMY dataset is publicly available and can be downloaded in monthly files from the following link: https://zenodo.org/records/14632036. The code used to construct this dataset is available upon request.

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